

Josiah Holder

Mechanical Engineer: product design, prototyping, and test

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SUMMARY

Mechanical engineer with 3.5 years developing agricultural sensor hardware from concept through production. Designs in Creo and Fusion 360 for cost and manufacturability, builds the test systems and analysis that prove a design meets spec, and works directly with farmers, dealers, and suppliers.

SKILLS

Design and CAD: Creo, Fusion 360, DFM, GD&T drawings, tolerance stack-up, drawing and model revision control

Manufacturing: Injection molding, SLS, machining, sheet cutting, overmolding, supplier quoting, end-of-line testing

Prototyping: FDM and SLS 3D printing, rapid prototyping, custom lab and field fixtures

Test and analysis: Test stand design, fluid and pneumatic rigs, thermal chamber testing, DAQ, CPK and PPK, Python, Julia, C#, introductory CFD

EXPERIENCE

Precision Planting: R&D Engineer

Jan 2025 - Present, Fargo, ND

- Developed three mechanical designs for an optical seed-flow sensor (injection molded, SLS printed for production, machined, sheet cut), quoted them with multiple manufacturers, and chose the lowest-cost option with no loss in usability
- Manage CAD, models, and drawings from prototype into production, including revisions, with flatness and surface finish callouts, ordinate dimensioning, and tolerance stack-up control
- Cut sensor-to-sensor variance about 3x by tightening the design to reduce tolerance stack-up
- Designed the validation system (test stand, control software, data pipeline, analysis) that proved the sensor met its $\pm 2\%$ requirement, with 95% of units within $\pm 2\%$ of the mean; extended measurement range about 80%
- Ran 50+ sensors on the bench and analyzed data from 250+ installed on farmer equipment over two seasons
- Built fluid and pneumatic test rigs (VFDs, blowers, pumps, flow meters, pressure sensors, venturi conveying) and ran thermal chamber environmental testing
- Designed the mechanical side of an overmolded connector assembly that houses a circuit board (entering production this winter), plus adapter harnesses and custom lab and field harnesses
- Built a Python STEP assembly viewer, packaged as a standalone app, for product support, saving an estimated 4+ engineering hours per product
- Visit farms and dealers to gather feedback, and work with suppliers and contract manufacturers on cost and end-of-line testing

Appareo Systems (Intelligent Ag): R&D Engineering Intern

Mar 2023 - Jan 2025, Fargo, ND

- Rescued a sensor program at risk of cancellation: took a mechanical design from non-functional to working, optimized it to hold accuracy across a harsh operating range, and built a heat map algorithm that improved accuracy from roughly $\pm 60\text{--}80\%$ to $\pm 5\%$
- Removed a calibration step that skewed the reference value and corrected for product variation across materials
- Owned lab and field testing, test stands, and data analysis for the project, at a full engineer's workload

Faul Building and Repair: Foreman

Summer 2021, Fargo, ND

- Led a crew of up to 8 on demolition, framing, and repair jobs; handled scheduling and served as the crew's contact with the owner and customers

Co-founder, small business

2024 - Present, Fargo, ND

- Building low-cost, easy-to-use technical solutions for farmers

EDUCATION

North Dakota State University: B.S. Mechanical Engineering

December 2024